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PROFILE

- Experienced wet-lab **protein biochemist**, evolved into a **computational designer**.
- Passionate about solving complex **biological puzzles** by combining structural and experimental insights with **AI/ML pipelines** to design the next generation of functional proteins.

KEY SKILLS

- Computational Protein Design and AI/ML.
- Structural Biology & Protein Characterization.
- Data Science, Analytics & Infrastructure.
- R&D Project Management & Collaboration.
- Python, MongoDB, SQL, R, AWS, and basic Apache Spark for ML.

ANA R. CORREIA

Work Experience

SENIOR AI (PROTEIN) ENGINEER | SAMSARA ECO NSW, AUSTRALIA | Jul 2024 - Present

- Protein design and engineering of novel enzymes for plastic recycling applications.
- Benchmarking and optimization of in-house ML models, while proactively auditing and integrating cutting-edge, open-source AI/ML protein engineering tools into discovery platforms.
- Integrate ML predictions with structural, biophysical, and experimental data to accurately prioritize high-quality variants for validation.
- Collaborate on architecting the company's data model and building the database infrastructure, to enhance data accessibility between teams and improve integration across engineering pipelines.
- Bridge computational and wet-lab teams to drive rapid design-build-test-learn iterations for novel enzyme development.

SCIENTIST | AMGEN INC. Thousand Oaks, CA, USA | Apr 2018 - Sept 2022

- Integrated multidisciplinary teams focused on leveraging and structuring data for AI/ML applications.
- Project development and management.
- Protein engineering, development of purification and analytical characterization workflows for established and novel antibody-like formats (including bi- and multivalent biologics).
- Mentoring, cross-training and supervision of junior scientists in purification, data analysis, and high throughput sample tracking systems.

POSTDOC | CALTECH | Hoelz Lab Pasadena, CA, USA | Nov 2013 - Mar 2018

- Expression, purification, assembly and characterization of large macromolecular complexes.
- Structural determination of protein complex by X-ray crystallography.
- Expression and characterization of synthetic antibodies, for further use in crystallization assays.

POSTDOC | CNB-CSIC | Caston Lab Madrid, SPAIN | Aug 2011 - Oct 2013

- Production of virus in mammalian and fish cell lines. Purification of virus and viral like particles and subsequent biochemical and structural characterization (negative-stain and Cryo-EM).
- Viral ribonucleoprotein complexes extraction and characterization.

VISITING RESEARCHER | BONN UNIVERSITY | Voos Lab Bonn, GERMANY | Feb - Jul 2010

- In vivo (yeast cells) and ex vivo (isolated mitochondria) characterization of a mitochondrial protein biogenesis.

Education

PHD STUDENT | ITQB-NOVA | Gomes Lab Lisbon, PORTUGAL | 2006 - 2010

- Conformational and functional characterization of a mitochondrial protein using different biophysical techniques namely: fluorescence spectroscopy, differential scanning calorimetry, calorimetry, circular dichroism and NMR.

INTERN AND VISITING PHD STUDENT | NIMR-MRC London, UK | 2005 - 2009

- Research training on molecular biology and NMR.

BACHELOR'S DEGREE, BIOCHEMISTRY, 2005 | UNIVERSITY OF LISBON (FCUL)

LANGUAGES

- English
- Spanish
- Portuguese (native)

HOBBIES & INTERESTS

- Hiking
- Scuba Diving
- Ocean Conservancy
- Mountain Biking
- Traveling

Selected Additional Training & Certifications

- **MicroMasters Programs MITx – Statistics and Data Science (2025).**
- Python 3 Programming by University of Michigan on Coursera (2022).
- Applied Machine Learning in Python by University of Michigan on Coursera (2023).
- Machine Learning Specialization by Stanford University & DeepLearning.AI on Coursera (2023).
- Deep Learning Specialization by DeepLearning.AI on Coursera (2023).
- Data Science Specialization by Johns Hopkins University on Coursera (8 courses from the specialization including R programming, 2021).
- Bioinformatics I, II by University of Toronto on Coursera (2023).
- Algorithms for DNA sequencing by Johns Hopkins University on Coursera (2023).
- SQL for Data Science by University of California, Davis on Coursera (2023).
- Apache Spark for Data Engineering and Machine Learning by IBM on edX (2023)

Proactively advancing my expertise in computational biology and data science to stay at the forefront of emerging industry methodologies and technologies.

Publications (8 as a first author, *)

- Utility of physiologically based pharmacokinetic modeling to predict inter-antibody variability in monoclonal antibody pharmacokinetics in mice. (2023) **mAbs**, 15(1): 2263926.
- Development of in silico models to predict viscosity and mouse clearance using a comprehensive analytical data set collected on 83 scaffold-consistent monoclonal antibodies (2023) **mAbs**, 15(1):2256745.
- Architecture of the cytoplasmic face of the nuclear pore (2022) **Science**, 376 (6598):1174-1192.
- Enhancing the Prefusion Conformational Stability of SARS-CoV-2 Spike Protein Through Structure-Guided Design (2021) **Front. Immunol.**, 12, article 660198.
- Rational selection of building blocks for the assembly of bispecific antibodies (2020) **mAbs**, 13 (1), e1870058.
- Structural and functional analysis of mRNA export regulation by the nuclear pore complex (2018) **Nat. Commun.**, 9 (1):2319-2338. *
- A new MR-SAD algorithm for the automatic building of protein models from low-resolution X-ray data and a poor starting model (2018) **IUCrJ.**, 5 (Prt 2):166-171.
- Nuclear pores. Architecture of the nuclear pore complex coat (2015) **Science**, 347, 1148-52. *
- Probing the kinetic stabilities of Friedreich's Ataxia clinical variants using a solid phase GroEL chaperonin capture platform (2014) **Biomolecules**, 4 (4): 956-79. *
- Revertants, low temperature, and correctors reveal the mechanism of F508del-CTFR rescue by VX-809 and suggest multiple agents for full correction (2013) **Chem. Biol.**, 20 (7):943-55.
- Iron binding activity in yeast frataxin entails a trade off with stability in the a1/b1 acidic ridge region (2010) **Biochem. J.**, 426(2):197-203. *
- No evidence of direct binding between ursodeoxycholic acid and the p53 DNA-binding domain (2010) **Biosci Rep.**, 30(5):359-64. *
- The conserved Trp-155 in human frataxin as a hotspot for oxidative stress related chemical modifications (2009) **BBRC**, 390(3):1007-11. *
- Dynamics, stability and iron-binding activity of frataxin clinical mutants (2008) **FEBS J.**, 275, 3680-3690. *
- Conformational stability of human frataxin and effect of Friedreich's ataxia related mutations on Protein Folding (2006) **Biochem. J.**, 398, 605-611. *